

*Bank of Korea International Conference*

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# Fiscal Influences on Inflation

By Robert Barro and Francesco Bianchi

Discussion:  
Saki Bigio

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# Why This Paper Matters to Me

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I lived through Peru's hyperinflation

Was taught fiscal-monetary theory (e.g, Sargent & Wallace)

...but inflationary finance, absent from core NK (DSGE) models

Francesco: been pushing the frontier, paper no exception

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# Barro-Bianchi

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OECD country panel

Finding: strong link between  $\pi$  and G

- with debt maturity and debt-level weakening link

Interpretation: **evidence of Fiscal Theory of Price Level (FTPL)**

- Key claim: inflation **automatically** erodes public debt, enabling solvency

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# Inflationary Finance $\neq$ FTPL

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Data show inflation indeed eroded public debt

- clear evidence of **inflationary finance**

But,... paper presents this as **evidence of FTPL**

I disagree: evidence **neither in favor nor against FTPL**

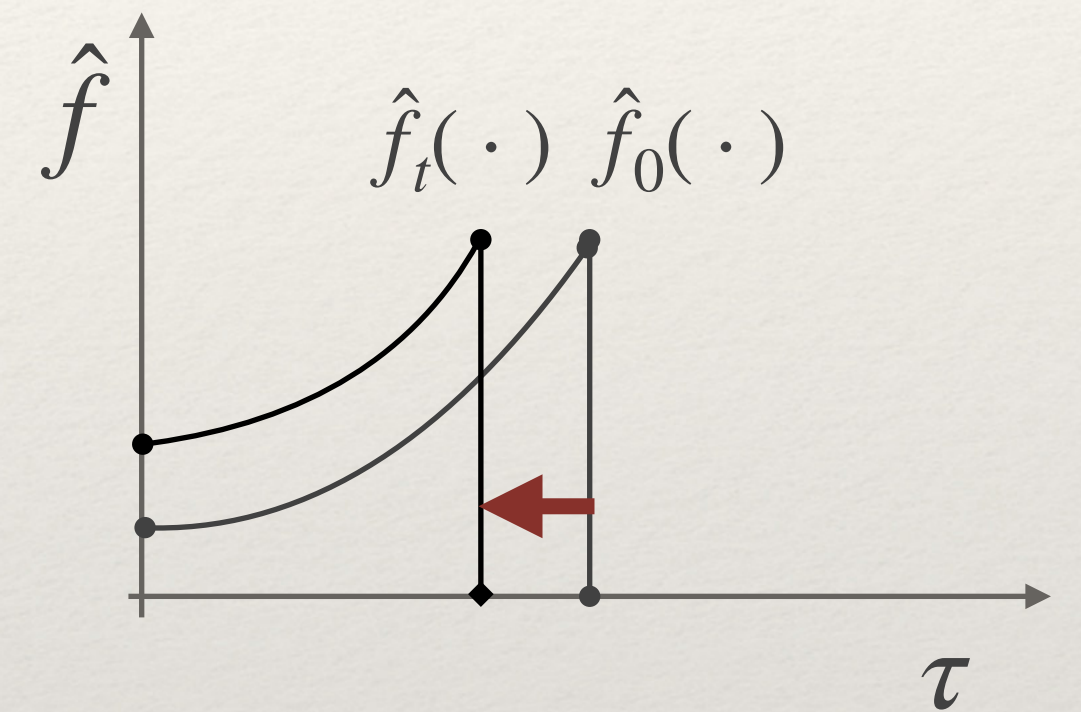
**Risky interpretation:** may conceal other risks

# A different look at the Gov's Budget

Deterministic  $t \in [0, \infty]$ , big shock at  $t = 0$

Nominal payment schedule  $\hat{f}_t(\tau)$ ,  $\tau$  maturity,  $\tau^{max} \leq T$

$$\dot{\hat{f}}_t = \frac{\partial \hat{f}}{\partial \tau}, \quad \hat{f}_t(0^-) = 0$$



Debt contracted at  $t \geq 0$  is  $b_t$

$$\dot{b}_t = rb_t + g_t - \theta_t + \frac{\hat{f}_t(0)}{P_t}$$

 expiring schedule

# Debt Erosion

Real payment schedule satisfies:

$$f_t(\tau) = f_0(\tau + t) \cdot e^{-\int_0^t \pi(s) ds} \rightarrow \text{erotion in that time}$$

*expired time*

Present value of government budget:

$$\Gamma + \int_0^T e^{-\int_0^t r(s) ds} \cdot f_0(t) \cdot e^{-\int_0^t \pi(s) ds} dt = e^{-\int_0^T r(s) ds} \cdot b_T$$

present value primary deficits:

$$\Gamma = \int_0^T e^{-\int_0^t r(s) ds} \cdot (g_t - \theta_t) dt$$

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# The Barro-Bianchi Formula

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Clean comparative static (constant inflation)

$$\frac{d\bar{\pi}}{d\Gamma} = \frac{1}{\mathcal{F} \cdot \mathcal{T}}$$

where

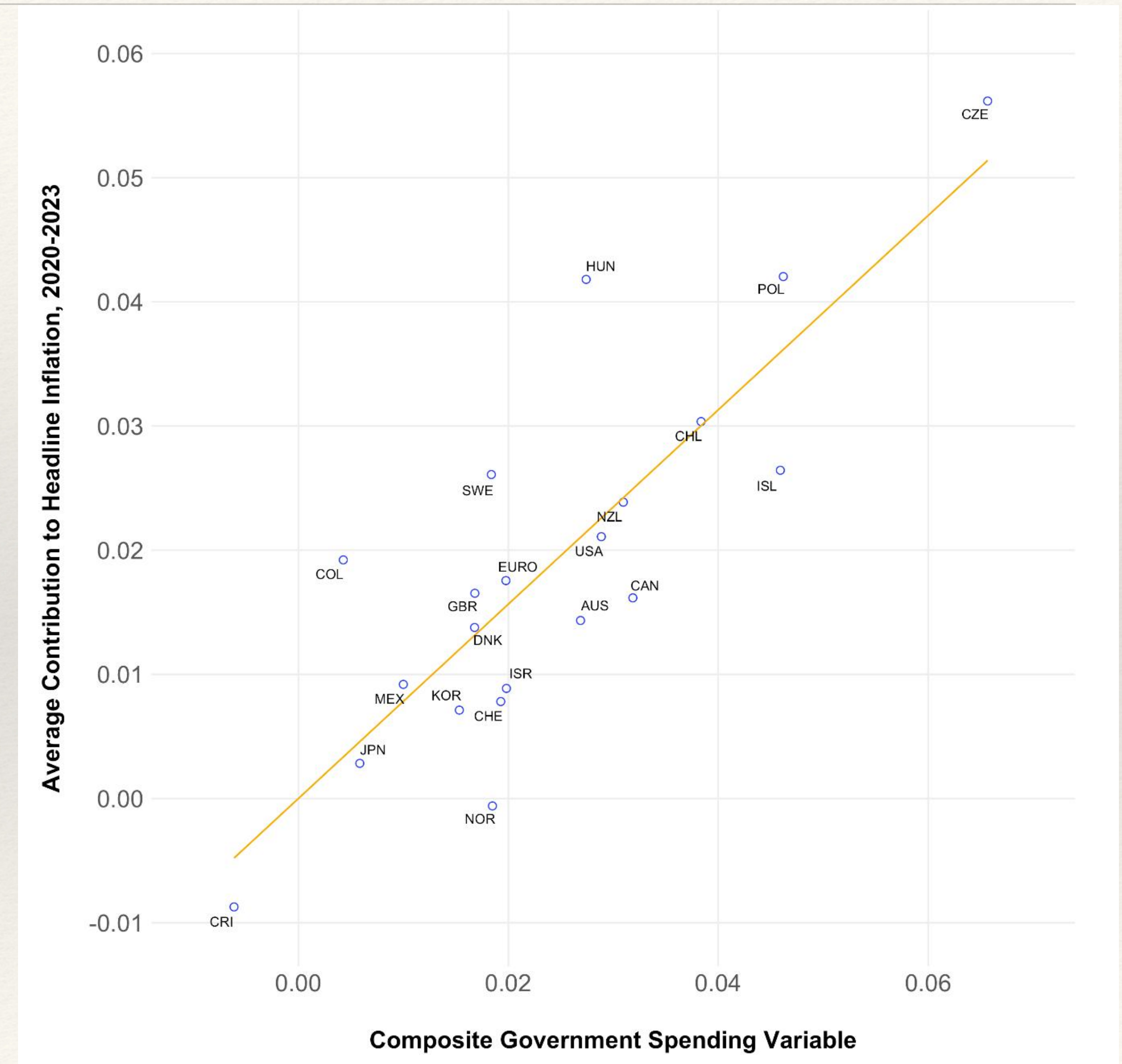
$$\mathcal{F} = \int_0^T e^{-\int_0^\tau i(s) ds} f_0(\tau) d\tau \quad \text{and} \quad \mathcal{T} = \int_0^T \tau \cdot \omega(\tau) d\tau$$

$\omega(\tau)$ : share at maturity

- Inflation inversely to market value debt and Macaulay maturity

# Theory meets Data

$$d\bar{\pi} = 1 \times \frac{d\Gamma}{\mathcal{F} \cdot \mathcal{I}}$$



Main Regression

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# Budget Constraints Are Not FTPL

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Results validate **inflationary finance**—not FTPL

FTPL is about **equilibrium selection**, not budget constraints

- Budget identities holds in **all** models: NK, monetarist, FTPL

Cochrane (Ch. 22, 2023):

“Equivalence is equivalence... The government debt valuation equation is an equilibrium condition in all of these models (NK, FTPL, Monetarist).”

# NK vs. FTPL

## NK (active MP)

$$\dot{x}_t = (\phi - 1) \cdot \pi_t - \rho + u_t$$

$$\dot{\pi}_t = \rho\pi_t - \kappa x_t$$

$$\dot{b}_t = i_t - \pi_t - \rho - \gamma b_t + \psi_t$$



decoupled system

2nd order ODE: (-) boundary

## FTPL (active FP)

$$\dot{x}_t = (\cancel{\phi} - 1) \cdot \pi_t - \rho + \tilde{u}_t$$

$$\dot{\pi}_t = \rho\pi_t - \kappa x_t$$

$$\dot{b}_t = i_t - \pi_t - \rho - \cancel{\gamma} b_t + \tilde{\psi}_t$$

Selection: Taylor principle = blow up of inflation

Fiscal shocks decoupled

Selection: no inflation = blow up budget

Fiscal shocks  $\rightarrow$  jump in  $\pi_0$

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# What We Don't Learn

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1) No **behavior features** → no causal evidence

2) Doesn't explain **inflation**—only that it helped

Example: **accommodative monetary policy opened up fiscal space**

Case: Costa Rica had stringent fiscal rule

→ no inflation and low  $G$

→ but fiscal rule counter to FTPL

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# Also...

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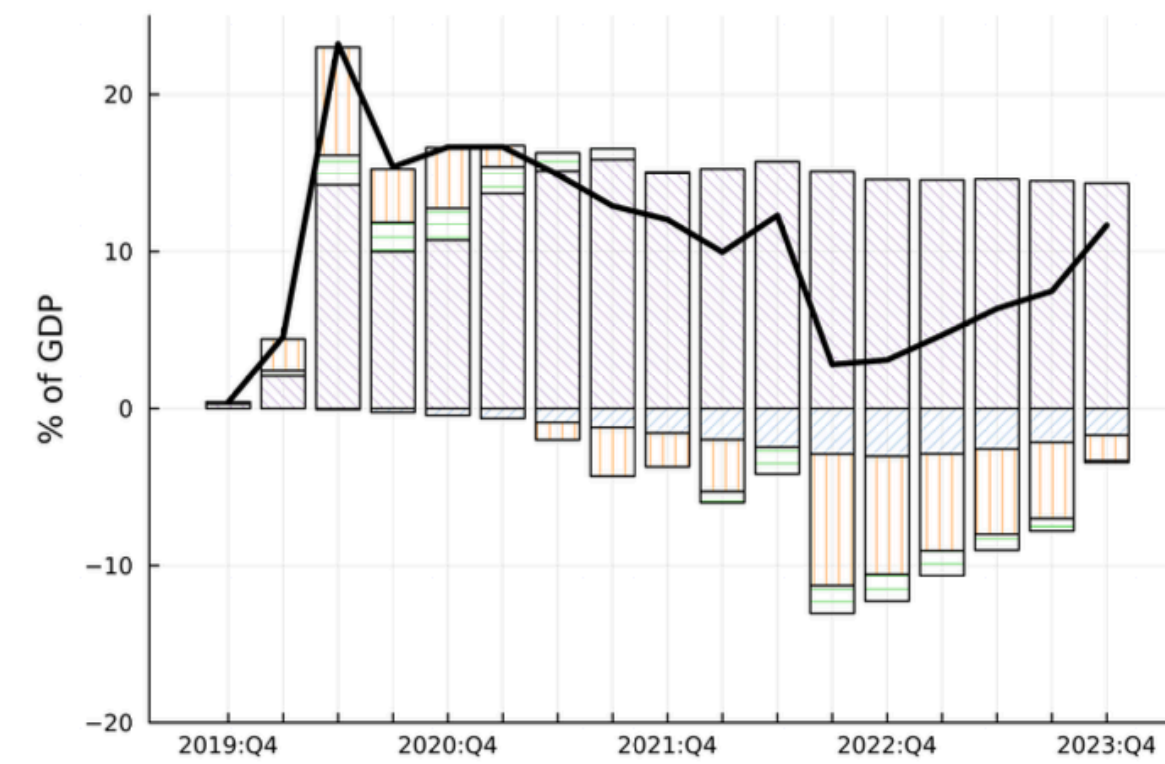
Incomplete picture of budget finance

Missing: bond prices (input in Barro-Bianchi formula)

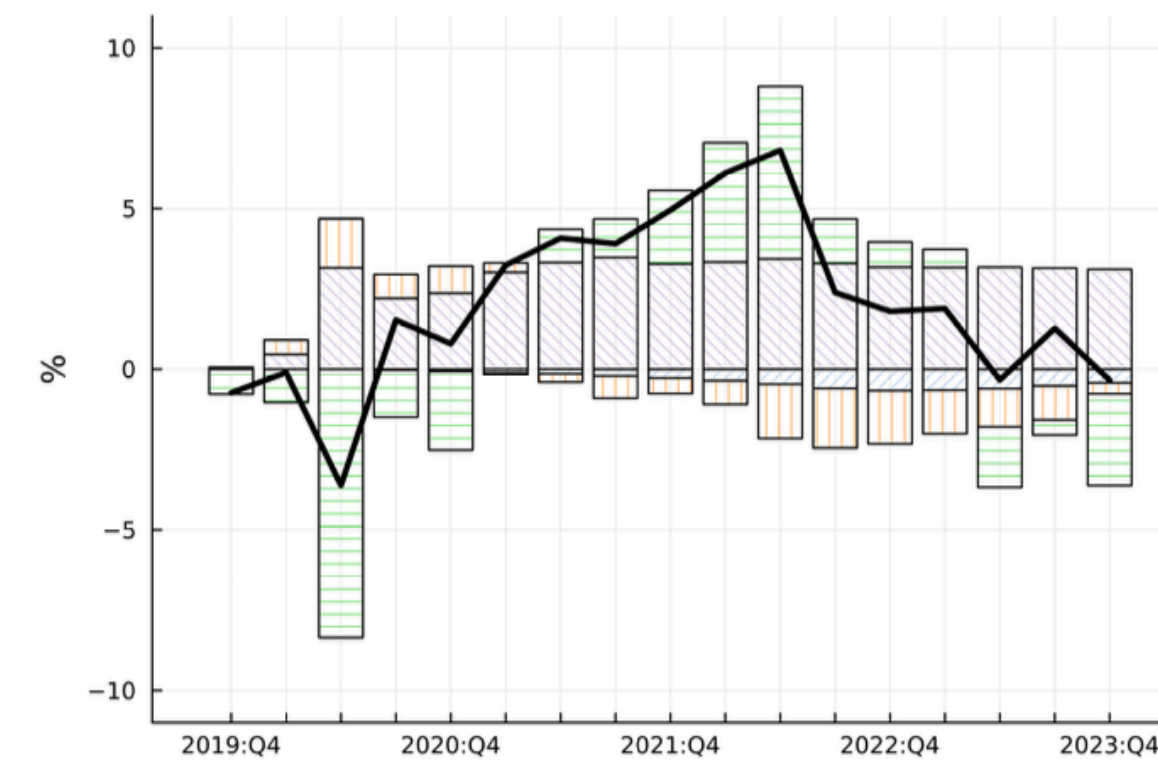
$$\mathcal{F} = \int_0^T \exp \left( - \int_0^\tau i(s) ds \right) f_0(\tau) d\tau$$

Financing: joint composite of inflation and yields

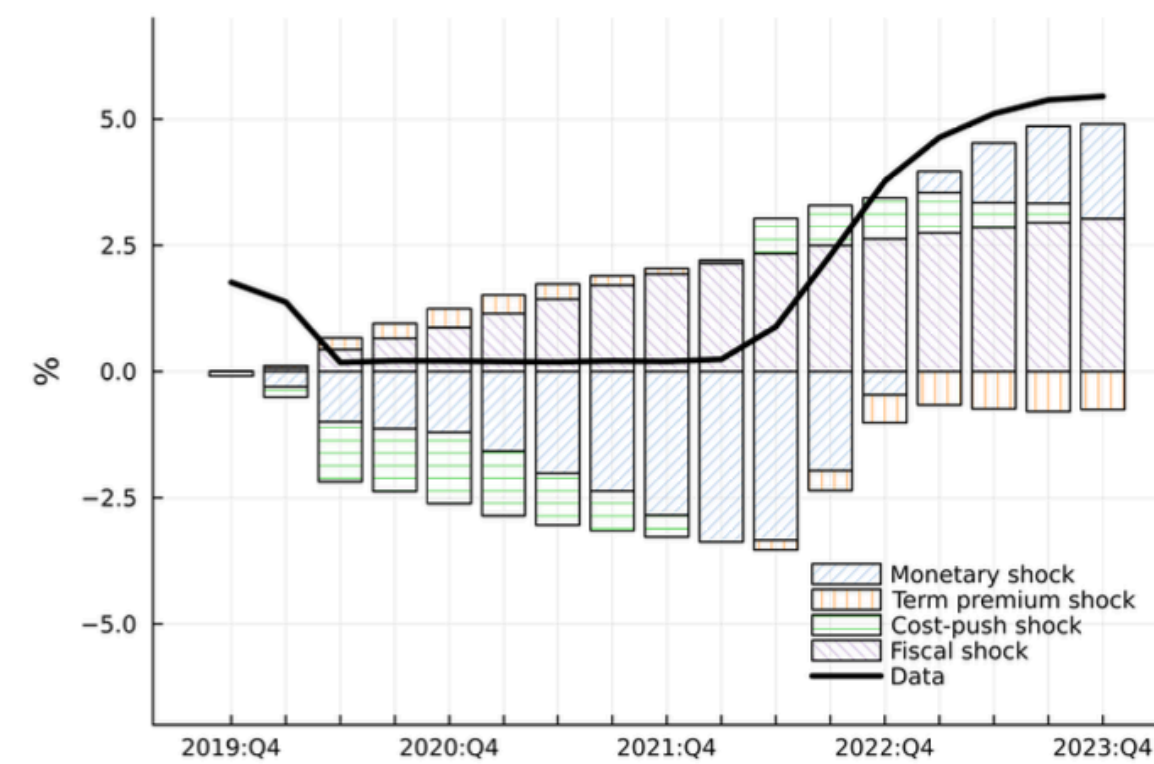
# A decomposition w/o FTPL



(a) Government Debt to GDP



(b) Inflation



(c) Federal Funds Rate

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# The Risk of FTPL interpretation

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FTPL interpretation as "automatic" debt stabilizer **may conceal greater risk**

I fear **American roll-over crisis**

- **FTPL says its ok if FED stays put**

Sustainability **may need accomodation** to avoid more inflation if **roll-over crisis**

- **but....inflation also undermines credibility and risks roll-over crisis**

Need richer model: explicit off-equilibrium behavior

- **complicate risks with international payments, foreign reserves, etc.**

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# Conclusion

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Impressive empirical work: simple formula w/ impressive fit

- convincing: global inflationary finance during COVID-era

FTPL?

- need testable implication of **equilibrium selection**, not budget identity

Warning: roll-over crisis may be across the corner

- risk should be core discussion, FTPL conceals this